

Yao Ming (Brian) Chan

PERSONAL INFORMATION

Email: brichan17@gmail.com

Date of birth: November 17th 1999

Place of birth: Ipoh, Malaysia

Nationality: Australian

Website: <https://brichan17.github.io>

EDUCATION

10/2023–Now **DPhil Mathematics**
University of Oxford
Current thesis title: On the Cuntz semigroups of non-simple C^* -algebras
Supervisor: Professor Stuart White

03/2021–11/2022 **MSc. Master of Science (Mathematics and Statistics)**
University of Melbourne
Major: Pure Mathematics
Thesis: Wedge product matrices and applications

Wedge product matrices are developed as a generalisation to the determinant of a square matrix with entries in a commutative ring. In tandem with a method Robert Steinberg used to prove a variant of the Bruhat decomposition, the two ideas were applied to various problems in linear algebra. Notably, they were used to generalise the eigenvector-eigenvalue identity and the notion of a quasideterminant in a non-commutative ring. They were also applied to the construction of representatives of two different matrix orbit spaces.

Supervisor: Professor Arun Ram

03/2018–11/2020 **BSc. Bachelor of Science**
University of Melbourne
Major: Mathematics and Statistics
Specialisation: Pure Mathematics

TEACHING

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| 2025 | Class tutor for Functional Analysis I B4.1 |
| | Organised and taught one set of classes (four classes total). Also played the role of the teaching assistant and marked the students' submitted work before each class. |
| 2025 | Teaching assistant for Functional Analysis II B4.2 |
| | Prepared for and taught one set of classes (four classes total) with the class tutor. Also marked the students' submitted work before each class. |
| 2023, 2024 | Teaching assistant for Functional Analysis I B4.1 |
| | Prepared for and taught one set of classes (four classes total) with the class tutor. Also marked the students' submitted work before each class. |

ACHIEVEMENTS

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| 2021, 2022 | Mathematics and Statistics Masters Scholarship |
| | Three payments of \$2000 corresponding to the first three semesters of the Master of Science degree in recognition of consistent academic achievement throughout the degree. |
| 2018, 2019 | Dean's Honours List |
| | Awarded for an average mark which lies in the top 3% of students in the Bachelor of Science degree. |

PREPRINTS AND PUBLICATIONS

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| 1. | Y. Chan. <i>Wedge product matrices and orbits of principal congruence subgroups</i> , Lin Alg App. 696 , 2024, Pages 1-28. |
| 2. | Y. Chan. <i>The Cuntz semigroup of a unital graph C^*-algebra</i> , Currently in preparation. |

TALKS DELIVERED

25/06/2026	The Cuntz semigroup of a unital graph C*-algebra (25 minutes), UK Operator Algebras Conference, Sabhal Mòr Ostaig, Isle of Skye.
08/06/2026	The Cuntz semigroup of a unital graph C*-algebra (50 minutes), Junior Functional Analysis Seminar, University of Oxford.
20/03/2026	The Cuntz-Krieger relations in the context of the Cuntz semigroup (15 minutes), Operator algebras in the south, Cardiff University.
11/09/2025	Purely large extensions with purely infinite quotients (20 minutes), Topological phenomena in the Cuntz semigroup, University of Oxford.
21/07/2025	Cuntz semigroups and graph C*-algebras (3 minutes), YMC*A, University of Southern Denmark, Odense.
04/07/2025	Cuntz semigroups and graph C*-algebras (5 minutes), UK Operator Algebras Conference, Queen's University Belfast.
03/04/2025	Nuclear dimension, extensions and graph C*-algebras (20 minutes), Young Functional Analysts' Workshop, University of Glasgow.
12/06/2024	Nuclear dimension and graph C*-algebras (5 minutes), UK Operator Algebras Conference, Newcastle University, Newcastle upon Tyne.

SKILLS

	Coding experience with Python
2021	Quantum chemistry project for the subject COMP90072 Used NumPy to implement the Hückel method in order to model the energy levels, electron densities and bond orders of conjugated organic molecules. The Hartree-Fock method was also implemented to compute the orbital energies of small atoms with occupied 1s and 2s orbitals and diatomic molecules comprised of these atoms.

LANGUAGES

English (Native)

PROFESSIONAL INVOLVEMENT

2025-2026 **Co-organiser** for the “Young Mathematicians in C^* -algebras” conference

REFERENCES

Professor Arun Ram (MSc supervisor)

Institute: University of Melbourne

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